Constraining Turbulence in the Intracluster Medium with the Radio Sunyaev Zeldovich Effect

by

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CURRENT DEGREES

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Abstract

Gas motions. SZ effect Pressure fluctuations. Stable methodology. modedling. simulation and some interferometry.

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For F.N.E.. May she live forever.

Contents

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List of Abbreviations and Symbols

Abbreviations	
AC	Alternating Current
AFM	Atomic Force Microscopy/Microscope
etc.	*etc.*

Symbols	
$\hat{\rho}$	Density operator
etc.	*etc.*

Chapter 1

Scientific landscape

> "If you wish to create a Universe from scratch, the final ingredient would be clusters of galaxies."
>
> — [Afiq Hamid 2025]

1.1 Galaxy Clusters

It has been said that mass and time are among the most fundamental parameters in all of astrophysical science. This thesis is principally an exploration of the former through the study of the most massive gravitationally bound structures in the Universe: galaxy clusters.

Galaxy clusters occupy a unique position within the hierarchy of astrophysical structures. They represent the endpoint of cosmic structure formation, populating the nodes that connect the filaments of the cosmic web [ref]. The matter composition of galaxy clusters is dominated by a hitherto enigmatic, collisionless dark matter (DM) component ($\sim 80\%$), while the remaining baryonic constituents are made up of the intracluster medium (ICM), a diffuse and turbulent plasma ($\sim 15\%$), with the luminous member galaxies — themselves home to most of the stars, planets, and people (human or otherwise) within the cluster — make up the smallest residual ($\sim 5\%$) [ref].

Structurally the DM component dictates the overall potential well, setting the stage for all of the baryonic interactions [ref]. At the base of this immense potential well sits the largest galaxy known as the brightest cluster galaxy (BCG), that accounts for 0.5–2% of the cluster baryons. BCGs commonly hosts an active galactic nucleus (AGN), a powerful source of energetic feedback attributed to super massive black hole (SMBH) activity as the origin [ref]. Understanding AGN physics is of broad interest in modern astrophysics, spanning diverse research areas from exotic accretion processes to high powered jet emission [ref], [ref], and while not the primary focus of this work, AGN feedback has major implications towards the stirring of gas motions within the ionised ICM.

Permeating the space between the member galaxies in a cluster is the ICM. It is a hot ($T \sim 10^7$–10^8 K), diffuse ($n_e \sim 10^{-3}$–10^{-1} cm^{-3}), optically thin, volume-filling plasma. The ICM is threaded by a weak but dynamically relevant magnetic field ($\sim$ 1–10 μG), detectable via Faraday rotation measures [ref]. As the primary atmosphere of the cluster, the ICM plays an important role in various dynamical processes therein.

- ***AGN thermostat*** The ICM regulates the cooling of gas towards the cluster core — a process that, if left unchecked, would produce excess star formation rates (SFR) in the BCG. In the absence of a heating mechanism, the dense core would radiate energy efficiently and cool on timescales far shorter than the cluster age, driving a so-called cooling flow [ref]. This cooling is now understood to be moderated by a self-regulating feedback loop: as gas cools and accretes onto the BCG it fuels the central AGN, which injects energy back into the ICM via jets and lobes, suppressing further cooling of the core regions [ref]. The ICM therefore acts as both the fuel supply and calorimeter of this baryonic cycle.

- ***Star formation in member galaxies*** The ICM plays an influential role on the evolutionary life-cycle tracks and SFR of its member galaxies through quenching mechanisms and ram-pressure stripping (RPS). As the satellite galaxies fall into the cluster potential well, contact with the ICM removes internal cold gas reservoirs from the interstellar medium (ISM) of the galaxies, quenching star formation on relatively short timescales ($t \sim 0.3$ Gyr) [ref]. This is most evident on close approach to the core and manifests as the stripped tails of jellyfish galaxies [ref]. Conversely, on longer timescales ($t \sim 3$ Gyr) strangulation targets the outer hot gas halo and prevents new gas from cooling [ref], [ref].

- ***Cluster mass bias*** Cluster mass measurements can be used as cosmological probes of the large scale matter distribution of the Universe constraining several important parameters of the ΛCDM cosmology model [ref]. Since the DM component cannot be directly observed by any present means, the ICM provides a useful observational marker to constrain cluster masses via complementary multi wavelength approaches. However, the approaches are heavily dependent on the assumption that the ICM gas exists in a state of quiescent hydrostatic equilibrium (HE) within the DM potential well [ref],[ref].

The observational leverage acquired on all three of the aforementioned processes is mediated through how the ICM manifests across the electromagnetic spectrum. The next section details the primary observational windows through which the ICM is studied in order to derive constraints on cluster mass, and sets the groundwork for the key physical quantities and input assumptions that underpin the contents of the later chapters.

1.2 Observables

Optical light as the oldest astronomical window was used to initially identify and catalog the first galaxy groups and clusters [Abell 1958]. It remains a viable way to evaluate the matter content of galaxy clusters via gravitational lensing [Kneib & Natarajan 2011] and velocity dispersion of member galaxies [Girardi et al. 1998]. However, the windows of most relevance toward understanding the thermodynamic properties of the ICM exist on opposite ends of the electromagnetic spectrum: namely, the X-ray and millimeter-wave radio.

1.2.1 X-ray emission

The X-ray window played a vital role in the initial discovery and characterisation of the ICM. The ICM of galaxy clusters remain among the brightest X-ray sources in the sky to date [ref]. The emission is predominantly produced via thermal bremsstrahlung (free-free emission), in which electrons are deflected by ions and radiate photons in the process. The X-ray surface brightness S_{X} of the ICM electrons is proportional to the line-of-sight (LOS) emission integral:

$$S_{\mathrm{X}} \propto \int n_e n_p \Lambda(T)\,\mathrm{d}l \approx \int n_e^2 \Lambda(T)\,\mathrm{d}l, \tag{1.1}$$

where n_e and n_p are the electron and proton number densities respectively, $\Lambda(T)$ is the plasma cooling function. Since the emissivity scales as n_e^2, X-ray probes ICM gas density and in combination with grazing-incidence, total external reflection optics provides high resolution studies of the ICM core. However, this sensitivity can be a double-edged sword as the n_e^2 dependence means that S_{X} can be biased by an unknown gas clumping factor arising from inhomogeneity in the core [ref].

Since the sound crossing time (represented by the time it takes for pressure fluctuations to travel across a region in the ICM) within an Abell radius of 3 Mpc is only 0.17 - 0.33 of the Hubble time (the timescale of the cluster formation and evolution), it is generally assumed that the ICM will eventually settle into a state of approximate HE where pressure forces counter act against gravity:

$$\frac{1}{\rho_g}\frac{dP}{dr} = -\frac{d\phi}{dr} = -\frac{GM(<r)}{r^2}, \tag{1.2}$$

where $M(<r)$ is the total mass enclosed within the spherically symmetric radius r, ϕ is the gravitational potential, ρ_g is the gas density, and P is pressure. This balance of forces is the gas-phase expression of the clusters's broader virialized state and provides a foundational framework through which X-ray observations can be used to infer cluster mass. Idealised hydrodynamic-simulation tests of the HE assumption show some useful agreement with true cluster mass, with RMS scatter of 15 % within the central few Mpc but growing mildly with radius, and can balloon locally to a factor of two in the presence of shocks or unrelaxed substructure [Schindler 1996] [ref].

Another important concept that emerged from the X ray regime is the analytical β-model. By applying an approximation for a self-gravitating collisionless system to a radial model of gas distribution in equilibrium within the same potential [Cavaliere & Fusco-Femiano 1976] derived the analytic isothermal radial β model of the ICM electron number density:

$$n_{gas}(r) = n_0 \left[1 + \left(\frac{r}{r_c}\right)^2\right]^{-\frac{3}{2}\beta}, \tag{1.3}$$

where n_0 is the central gas number density, r_c is the radius of the cluster core and the index $\beta \sim 2/3$ is known as the canonical β value [ref]. The profile describes a flat central core transitioning to a power-law decline at large radii. Its broad applicability across the cluster population follows naturally from the self-similar nature of galaxy clusters. While modern high-resolution observations and hydrodynamical simulations have revealed limitations of the model — particularly in its inability to capture asphericities from complex merging and dynamically disturbed systems — it remains a robust and widely used first approximation for the ICM radial gas density distribution [ref].

1.2.2 Radio Sunyaev-Zel'dovich effect

The thermal Sunyaev–Zel'dovich effect (tSZ) is a secondary anisotropy of the cosmic microwave background (CMB) arising from the inverse Compton scattering of CMB photons by hot thermal electrons within the ICM [Sunyaev & Zel'dovich 1970], [Sunyaev & Zel'dovich 1972]. As CMB photons traverse the cluster, a small fraction are energetically upscattered, producing a characteristic frequency dependent spectral distortion given by:

$$\Delta I_\nu = y \frac{2h\nu^3}{c^2} \frac{xe^x}{(e^x-1)^2} \left[x\frac{e^x+1}{e^x-1} - 4\right], \tag{1.4}$$

where $x = h\nu/k_B T$ is the dimensionless frequency, $T = 2.725$ K is the CMB temperature, h is Planck's constant, and k_B is Boltzmann's constant. The amplitude of the distortion is set by the Compton y parameter, which is proportional to the LOS integral of the electron pressure:

$$y = \frac{\sigma_T}{m_e c^2} \int P_e dl, \tag{1.5}$$

where $P_e = n_e k_B T$ and σ_T is the Thomson cross section. The Compton y parameter therefore provides a direct in-situ measure of the integrated thermal pressure of the cluster along the LOS. A key advantage of the tSZ signal over S_X is its linear dependence on electron density making it more sensitive to the low-density outskirts of the cluster towards the virial radius. Coupled with a sufficiently wide field of view (FOV), the tSZ effect is therefore more suited to mapping the thermodynamic structure of the ICM on large scales and by shows promise in mapping the cosmic baryon distribution [Mroczkowski2019]. Figure 1.1 shows a multispectral image of the Coma cluster of galaxies combining both tSZ and S_X.

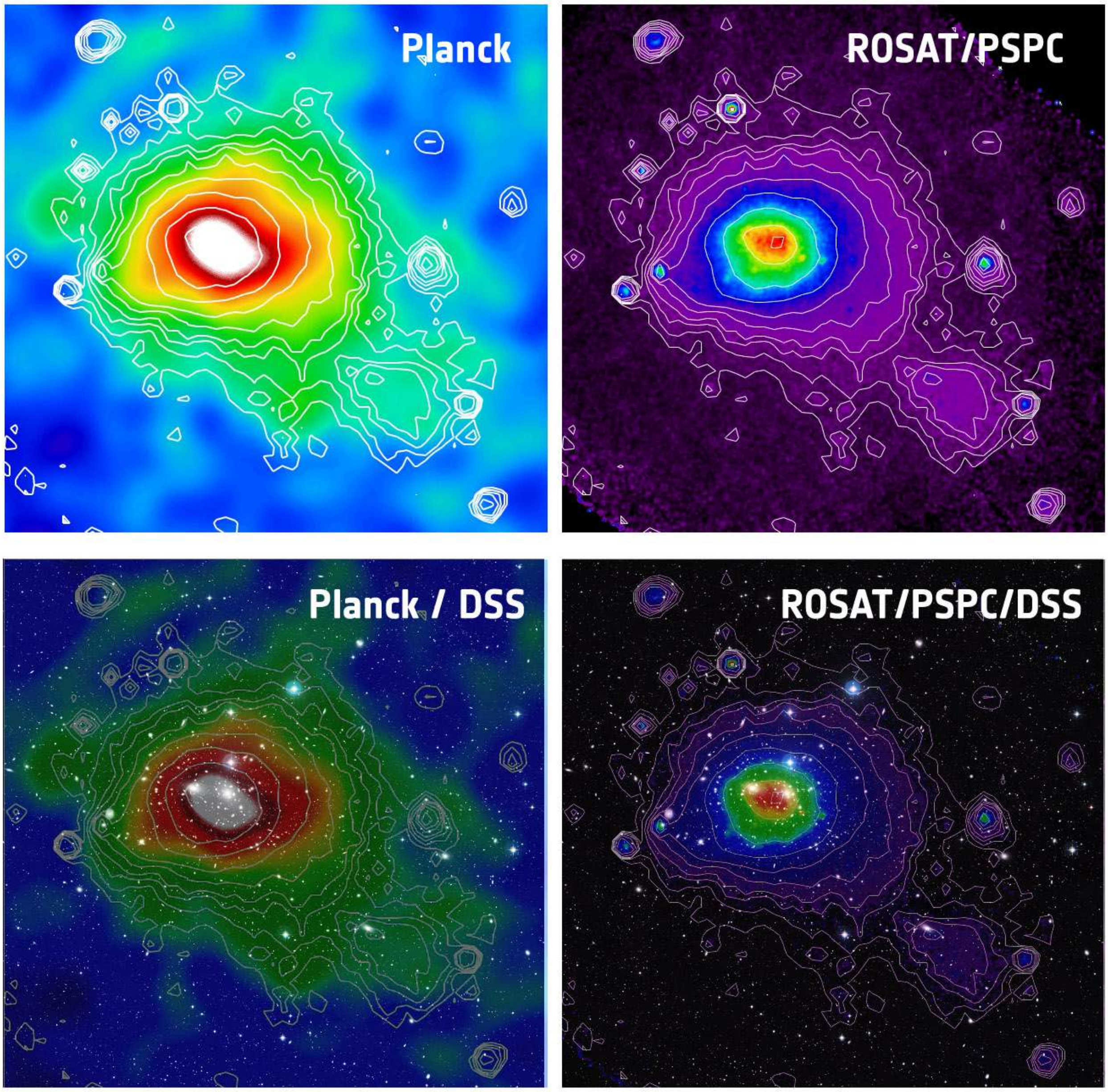


Figure 1.1: Multiwavelength composite of the Coma cluster (Abell 1656). Top left: Planck tSZ Compton y map. Top right: ROSAT soft X-ray surface brightness. In both upper panels the X-ray contours are superimposed on the tSZ image to illustrate the complementary spatial coverage of the two windows. The primary difference between the two regimes is that the tSZ emission is more spatially extended while the X-ray image is more centrally peaked. The bottom panels show the mult spectral image overlaid on the optical image, further revealing the member galaxy distribution. [ref] [ref].

1.3 ICM structure and non-thermal pressure

Notwithstanding the overview provided thus far, the ICM is in reality a complex and dynamically evolving system. The true gas conditions are in actuality far removed from the idealised equilibrium state assumed prescribed by 1.2. A truer conceptual picture — one favoured in the present literature — models the ICM as a stratified, multiphase fluid confined by the cluster potential but dynamically governed by a combination of thermal and non-thermal processes. [Donahue & Voit 2022][Battaglia et al. 2012]. This section briefly interrogates each of these ideas in further detail.

A fluid description of the ICM

The treatment of the ICM as a fluid is suitable when the mean free path λ_{mfp} of the constituent particles is much smaller compared to the characteristic scale length L of the system. For the bulk ICM, $\lambda_{\mathrm{mfp}} \approx 4.6$ kpc [ref], which is well below the scales of interest for cluster-wide gas dynamics — justifying the use of the Euler or Navier–Stokes equations as a fluid description [ref]. However, this fluid approximation is not universal. In the low-density outskirts of clusters, where the mean free path is much higher, and on small scales where kinetic plasma effects become important, a purely hydrodynamic description breaks down and kinetic or magnetohydrodynamic (MHD) formulisms are more appropriate [ref].

Stratification

Stratification implies that there is a radial ordering of the different thermodynamic properties — principally density, temperature, and entropy all under the simultaneous influence of gravity [ref]. The condition of stratification is characterized by the Brunt–Väisälä (BV) frequency:

$$N_{\mathrm{BV}}^2 = \frac{g}{\gamma}\frac{d}{dr}\ln\frac{P}{\rho_{\mathrm{g}}^{\gamma}}, \tag{1.6}$$

where g is the local gravitational acceleration, γ is the adiabatic index, P is the thermal pressure, ρ is the gas density, and r is the radial coordinate. The quantity P/ρ^{γ} is proportional to the specific entropy K, thus N_{BV}^2 also describes an entropy gradient. When $N_{\mathrm{BV}}^2 > 0$ — as is generally the case throughout the bulk ICM where entropy increases outward — the stratification is stable: a radially displaced gas parcel experiences a restoring buoyancy force and oscillates about its equilibrium position at frequency N_{BV} rather than convecting freely. This suppresses radial mixing while also supporting the propagation of internal gravity waves (g-modes) which are themselves a carriers of turbulent energy [ref]. There are some edge cases that exist beyond the purely hydrodynamic cases that can introduce exotic processes such as magneto thermal instabilities (MTI) which argue for conditions that could prove degenerative to these stable N^2 state [ref]. In this case what happens?

Multiphase structure

The ICM is also not characterised by a single thermal phase. Modern observations and simulations reveal a multiphase environment in which gas spans a wide range of temperatures — from cold molecular filaments at $T \lesssim 10^2$ K to the bulk hot phase at $T \sim 10^7$–10^8 K [ref]. This stands in sharp contrast to the traditional picture of a uniform hot atmosphere, and draws analogy to the multiphase structure of the more local ISM.

Cold phases and filaments in the ICM develop as a result of thermal instability and chaotic cold accretion (CCA) [ref]. The CCA process within cluster cores is analogous to precipitation: where the local cooling time drops below the free-fall time, allowing for gass to condense out of the hot phase and gravitationally accrete onto the center as cold filaments and clouds [ref]. A key theoretical challenge of understanding this turbulent process is trying to figure out how the cold filaments are not evaoprated back into the hot phase [ref]. The current candidates are a mix of anisotropic thermal conduction, unique magnetic field alignments, and heat flux buoyancy instabilities (HBI).

1.3.1 Non-thermal pressure

Within the stratified, multiphase fluid described of the ICM a range of dynamical phenomena occur that challenge the HE assumption. These challenges come in the form of bulk motion, turbulence, cosmic rays, and magnetic fields that all contribute non-thermal pressure components. In reality, the total pressure support within the ICM is not provided by thermal component alone:

$$P_{\text{tot}} = P_{\text{th}} + P_{\text{nt}} = P_{\text{th}} + \frac{1}{3}\rho_{\text{g}}\sigma_v^2, \tag{1.7}$$

where σ_v is the velocity dispersion of isotropically moving gas decomposable into radial and tangential components [ref]. Random turbulence and ordered gas motions contribute the most dominant part of P_{NT}, with cosmic rays and magnetic fields contributing only at the level of a few per cent [ref]. From comparisons between theory and observation, residual gas motions are estimated to contribute between $\sim 10\%$ at R_{500} and $\sim 30\%$ at R_{200} of the total pressure support, with this fraction increasing in dynamically disturbed systems [ref].

Constraining P_{nt} observationally is a non-trivial problem. Direct constraints are in principle accessible through X-ray spectroscopy, with velocity broadening and shifts of spectral emission line methods [ref][ref]. However, the spatial coverage of current X-ray spectroscopic instruments is limited, and the technique is restricted to relatively nearby, X-ray bright systems as a result of cosmological dimming. The radio SZ surface brightness fluctuation methods investigated in this work offer a spatially extended alternative, at the cost of a greater number of modeling assumptions — the characterization and interrogation of which are attempted throughout the next two chapters.

Astrophysical sources of ICM gas motions.

The non-thermal pressure within clusters is sustained via continuous injection of kinetic energy into the ICM. The injection originates from a hierarchy of physical drivers operating across a wide range of spatial scales [ref]. These can be broadly ordered from large to small as follows.

On the largest scales, the dominant energy source is gravitational — energy enters the system through the hierarchical assembly of large-scale structure through major cluster mergers and the continuous accretion of unvirialized material along cosmic web filaments [ref]. Major mergers between clusters of comparable mass drive powerful shocks and large-scale bulk flows that inject kinetic energy on scales approaching the virial radius, with the resulting turbulence taking of order a Gyr to equilibriate. Figure 1.2 presents a gallery of merging clusters to illustrate the phenomenon.

On intermediate scales, minor mergers and glancing interactions on the scale of galaxy groups drive sloshing motions that set the cluster atmosphere into coherent oscillations without fully disrupting the cool core [ref]. These produce cold fronts — sharp contact discontinuities between the displaced cool core gas and the surrounding hotter ICM — visible as edges in S_X [ref]. The sloshing flows are indicated to be subsonic corresponding to velocities of several hunder kilometres per second and mach numbers in the range of 0.3-0.5 [ref]. The infall of dark matter subhaloes further stirs the ICM on scales intermediate between the merger and core scales, contributing gaseous wakes from RPS and turbulent substructure [ref].

On the smallest scales, AGN feedback injects mechanical energy directly into the cluster core via jets and radio-mode outflows, inflating cavities visible in S_X and driving weak shocks that propagate outward into the surrounding ICM [ref]. The buoyantly rising cavities transfer most of their energy to the gas after crossing several pressure scale heights of the atmosphere [ref]. Additional contributions arise from plasma instabilities — including MTI and HBI — that operate in the weakly magnetised ICM and can drive small-scale anisotropic turbulent motions, particularly in the cluster outskirts [ref].

The cumulative effect of this multi-scale energy injection is a turbulent velocity field and fluctuations that permeates the ICM from the virial radius inward. In the next section we consider some of the fundamental physical attributes of the turbulence phenomena are explored.

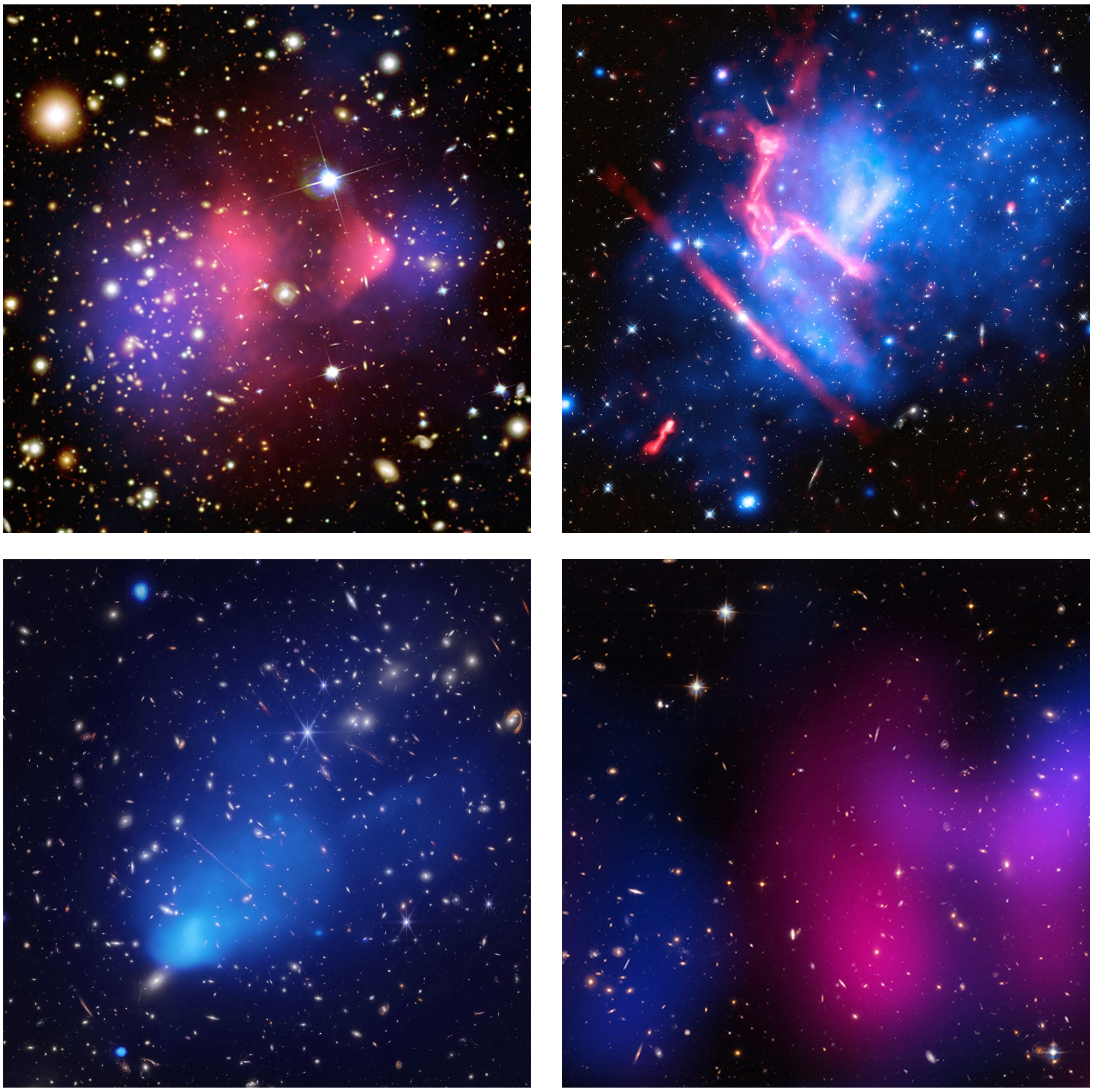


Figure 1.2: A gallery of dynamically disturbed galaxy clusters representing extreme violations of hydrostatic equilibrium. *Top left:* The Bullet Cluster (1E 0657-56), the archetypal binary merger. Chandra X-ray emission (pink) traces the shocked ICM stripped from the collisionless dark matter (blue, lensing reconstruction), providing the first direct observational evidence for dark matter.[ref]. *Top right:* MACS J0717.5+3745, the result of a quadruple cluster collision fed by a 13 Mpc cosmic web filament.Chandra X-ray (blue), VLA radio (red), and HST optical (background) reveal an extraordinarily complex thermodynamic structure [ref]. *Bottom left:* El Gordo (ACT-CL J0102-4915), the most massive known merging cluster at $z \sim 0.87$. Chandra X-ray emission (blue) is overlaid on a JWST infrared composite, with gravitational lensing arcs visible in the background field [ref]. *Bottom right:* The Musket Ball Cluster (DLSCL J0916.2+2951), a post-merger system caught further along in its evolution than the Bullet Cluster. The clear spatial offset between the hot gas (red) and dark matter (blue) illustrates the different collisional cross-sections of baryonic and dark matter [ref].

1.4 Turbulence theory

Turbulence is the study of irregular, chaotic fluid motion — one of the oldest unsolved problems in classical physics because of the non linear nature of its fundamental equations. In the terrestrial context it governs everything from the drag on aircraft wings to the mixing of cream in coffee. In the astrophysical context, the problem becomes even richer and more challenging: the characteristic scales incredibly vast with no well-defined boundaries, the fluid is self-gravitating, and the turbulence unfolds within a plasma intimately coupled to magnetic fields, and continuous energy injection from mergers and accretion. The ICM represents one of the most extreme turbulent environments in the Universe, and the true nature of its turbulent velocity field remains an open question in modern astrophysics.

The fundamental condition that allows for the onset turbulence is set by the Reynolds number:

$$\mathrm{Re} = \frac{u l_{\mathrm{inj}}}{\nu}, \tag{1.8}$$

where u is the characteristic velocity of the flow, l_{inj} is the driving scale, and ν is the kinematic viscosity of the fluid. Viscosity can be thought of as the amount of restiveness of the fluid. When $\mathrm{Re} \gg 1$, inertial forces dominate and viscous dissipation is negligible on the driving scale – a turbulent flow develops. For the ICM, estimates of the hydrodynamic Reynolds number is in the range $\mathrm{Re} \leq 100 - 10^3$ while the magnetic Reynolds number is high $\mathrm{Re}_{\mathrm{m}} \sim 10^{27} - 10^{29}$ [ref]. Thus if the ICM is observed on a purely fluid basis one would see a smooth slow pouring fluid while the magnetic fields would reveal a chaotic haywire configuration of field lines.

Turbulence energy cascade

The energy cascade is one of the fundamental concepts of turbulence theory [ref]. It describes a model of turbulence where energy is injected into the fluid at the driving scale l_{inj} corresponding to the scale of the dominant energy source, whether a major merger, sloshing motion, or AGN outflow. However, instead of dissipating immediately the energy is transferred progressively to smaller and smaller scales, with each generation of eddies feeding the next, until the viscous dissipation scale l_{diss} where the local Reynolds number approaches unity and viscous forces convert the remaining kinetic energy irreversibly into heat. The scale ranges between l_{inj} and l_{diss} constitutes the *inertial range*, within which the process of energy transfer occurs. For astrophysical scales this range typically spans many orders of magnitude [Armstrong et al., 1995].

Of the three scales of the energy cascade, the inertial range is of particular importance. It is within this range that the signature of specific turbulence regimes and physical models reveals itself. In the extreme environments of the ICM where MHD effects abound, shock fronts persist, and perhaps cases where both occur, resolving the characteristics of the inertial range is a paramount research objective.

However, limiting the scope to the classical hydrodynamic sense where the inertial range statistics are only dependent on the mean energy dissipation rate (assumption i) and energy transfer is local and takes place solely between eddies of roughly similar size (assumption ii), for an eddy of size l and characteristic velocity u_l thecharacteristic lifetime of an eddy of scale l is:

$$t_{\text{eddy}}(l) = \frac{l}{u_l}, \tag{1.9}$$

Within the inertial subrange, the assumption of a steady-state cascade requires that the rate at which energy enters a given scale equals the rate at which it leaves. This energy transfer rate, or cascade flux, is set by assumption (i) to be scale-independent and equal to ε. Dimensionally, it can be expressed as the kinetic energy per unit mass at scale l divided by the eddy turnover time,

$$\varepsilon \sim \frac{u_l^2}{t_{\text{eddy}}} \sim \frac{u_l^3}{l}, \tag{1.10}$$

Rearranging Equation (1.10) gives the characteristic velocity associated with eddies of size l,

$$u_l \sim (\varepsilon l)^{1/3}. \tag{1.11}$$

To express this result in spectral space, let $E(k)$ denote the omnidirectional/ shell-integrated energy spectrum, defined such that the integral $\int E(k)\,\mathrm{d}k$ is the energy within the range k and $k + \mathrm{d}k$. Since the cascade is local in scale (assumption ii) and a doubling of wavenumbers contributes a comparable share of energy, an eddy of size l corresponds to a band of wavenumbers centred on $k \sim 1/l$ with a bandwidth $\mathrm{d}k \sim k$. The total energy associated with eddies of size l can therefore be identified with the energy in the band:

$$u_l^2 \sim \int_k^{2k} E(k')\,\mathrm{d}k' \sim kE(k). \tag{1.12}$$

Substituting $l \sim k^{-1}$ into Equation (1.11) and squaring both sides yields:

$$u_l^2 \sim (\varepsilon l)^{2/3} \sim \varepsilon^{2/3} k^{-2/3}. \tag{1.13}$$

Joining Equations (1.12) and (1.13),

$$kE(k) \sim \varepsilon^{2/3} k^{-2/3}, \tag{1.14}$$

and finally solving for $E(k)$ yields:

$$E(k) = C_K\, \varepsilon^{2/3} k^{-5/3}, \tag{1.15}$$

where C_K is a universal non dimensional constant. This is the celebrated Kolmogorov -5/3 relation in turbulence statistics which finds applicability in diverse physical phenomena such as ocean and gulf stream currents, atmospheric boundaries, and various astrophysical plasmas [Grant 1962, Kaimal 1972, Goldstein 1995]. It is a foundational model which rests on several key assumptions — the turbulence is incompressible, statistically homogeneous, and isotropic. The 3D velocity power spectrum of this energy cascade is visualized in Figure 1.3.

Although the ICM does not in general satisfy the assumptions underlying Kolmogorov's derivation — the gas is compressible, the driving is anisotropic and intermittent, and the effective viscosity is poorly constrained due to the magnetised, weakly collisional nature of the ICM [ref] [ref] [ref] — the -5/3 relation nonetheless emerges as a useful template and well-motivated baseline to model turbulence. The Kolmogorov scaling has been encountered observationally in some astrophysical plasmas and even in the ICM of the Coma cluster but with some limitations [ref] [ref].

There is an analogy to be drawn with the radial gas distribution model: just as the β-model provides a tractable and physically motivated approximation to the global cluster emission despite not capturing every feature of the true surface brightness distribution, the Kolmogorov spectrum provides a tractable and physically motivated approximation to the inertial range statistics of the ICM despite not capturing every departure from ideality. In both cases, the value of the approximation lies not in its exactness but in its role as a controlled reference model against which deviations can be measured, and interpreted.

Additionally, the Kolmogorov cascade explored in the derivation describes the statistics of the *velocity* field — and in classical turbulence theory the velocity power spectrum is the primary diagnostic of the turbulent flow. However, the observational windows available for studying the ICM probe different thermodynamic quantities: X-ray surface brightness fluctuations are sensitive to density, while tSZ is a probe of pressure. An important question for ICM turbulence inference is therefore whether the density and pressure fluctuation power spectra carry sufficient information to map to the underlying velocity statistics. Numerical simulations have suggested that in the subsonic regime, such a mapping can exist in a surprisingly clean way [Gaspari 2014].

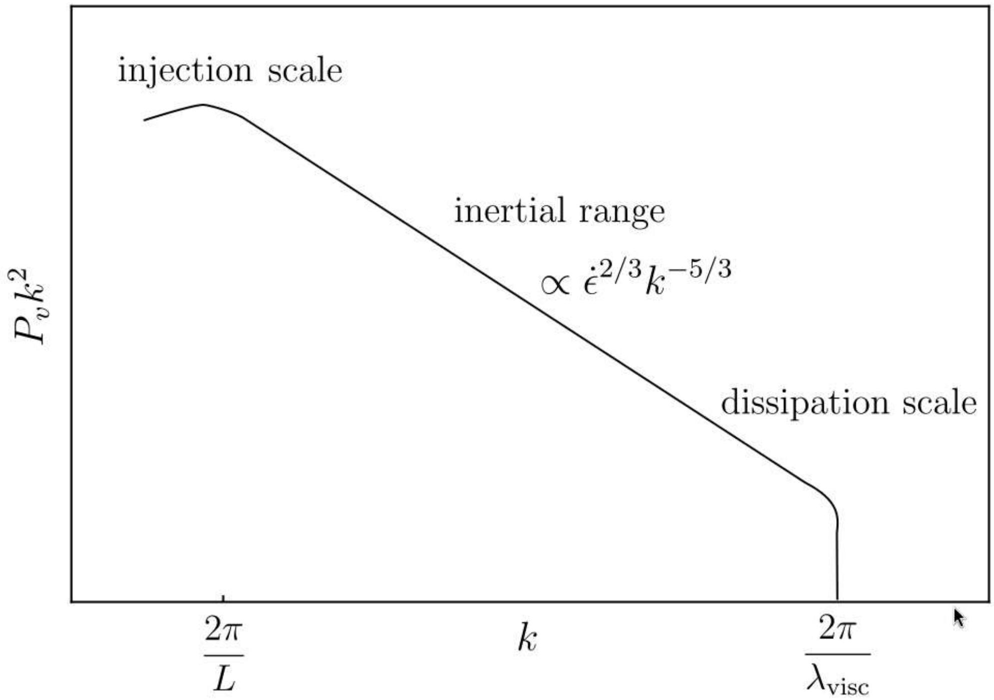


Figure 1.3: The turbulence cascade structure. The -5/3 value is that of omnidirectional spherical shells. If the velocity at a single point is measured over time and take a 1D Fourier transform, this would get the 1D longitudinal velocity spectrum (often denoted as $E_{11}(k_1)$). Interestingly, due to the mathematics of geometric projection, that 1D spectrum also scales as $k_1^{-5/3}$ in the inertial subrange, though with a slightly different C_K pre-factor.

1.5 Mass estimation and mass bias

The total mass of a galaxy cluster is one of its most fundamental parameters. Along with redshift, it bridges theoretical predictions with observational data, enabling the population of galaxy clusters to act as cosmological probes [Voit 2005, Kravtsov 2012, Allen 2011]. Since the mass is not directly observable; it must be inferred from ICM thermodynamic properties under a set of physical assumptions. The accuracy of those assumptions — and the systematic biases they introduce — therefore lies at the heart of cluster cosmology.

Concluding this introduction is an exploration of the principal frameworks for cluster mass reconstruction, their observational implementations, and the nature of the respective biases.

1.5.1 Generalized mass equation

The most physically complete description of the cluster mass profile that can be obtained is acquired by integrating the full Euler equation for an inviscid collisional gas over a spherical shell [Pratt 2019]. Inclusive of all of the dynamical terms, the total mass enclosed within radius r can be written as:

$$M_{\mathrm{tot}}(<r) = M_{\mathrm{therm}} + M_{\mathrm{rand}} + M_{\mathrm{rot}} + M_{\mathrm{cross}} + M_{\mathrm{stream}} + M_{\mathrm{accel}}, \tag{1.16}$$

where each term represents a distinct contribution to the pressure support against gravity [Lau2013]. The first term, M_{therm}, is the hydrostatic mass arising from isotropic thermal pressure. The second and third terms, M_{rand} and M_{rot}, account for the additional pressure support provided by turbulence and bulk gas motions, respectively. M_{cross}, M_{stream}, and M_{accel} capture contributions from cross-stream motions, bulk inflow or outflow, and local gas acceleration, all of which are more relevant during dynamically disturbed phases such as active mergers or rapid mass accretion.

Equation 1.16 is the definitive tool for numerical simulation. In a hydrodynamical simulation, all six components can be evaluated directly from three-dimensional velocity and pressure fields, providing complete knowledge of the mass budget at any radius and point of the cluster's assembly history. This makes it invaluable for quantifying the mass bias introduced by specific physical processes: sloshing cold fronts, AGN-driven cavities, merging substructure, and large-scale accretion shocks all leave distinct signatures across these terms [Rasia2006, Nagai2007, Lau2009, Nelson2014a, Biffi2016].

Despite its completeness, Equation 1.16 is inaccessible to direct observation. The three-dimensional velocity field cannot be confirmed from a single LOS vantage point, and even with next-generation X-ray spectroscopy [ref], spatially resolved velocity will remain limited to the inner cluster regions.

Hydrostatic mass equation

An alternative method of acquiring radial mass estimates is from rearranging Equation 1.2 to isolate $M(<r)$ and applying logarithmic derivatives yields:

$$M_{\rm HSE}(<r) = -\frac{k_B T(r)\, r}{\mu m_p G}\left[\frac{d\ln\rho_{\rm gas}(r)}{d\ln r} + \frac{d\ln T(r)}{d\ln r}\right], \tag{1.17}$$

Here $T(r)$ is the gas temperature, $\mu \approx 0.6$ is the mean molecular weight of ionised plasma, m_p is the proton mass, and G is the gravitational constant. This equation has been the workhorse of X-ray cluster cosmology and requires only two observationally accessible quantities where density profile $\rho_{\rm gas}(r)$, is obtained from the deprojection of X-ray surface brightness, and the temperature profile $T(r)$, obtained from spectral fitting. With the application of spatially resolved spectroscopy from *XMM-Newton* and *Chandra*, it has been used to derive continuous mass profiles for hundreds of clusters across a wide range of mass and redshift without requiring any assumptions about the structure of the cluster gravitational potential [Pointecouteau 2005, Vikhlinin 2006, Ettori 2013]. However, the weakness of this practice is also clear: it accounts only for thermal pressure support, and any residual non-thermal pressure contribution causes $M_{\rm HSE}$ to systematically underestimate the true mass.

A more complete estimate is obtained by factoring in the non-thermal pressure fraction:

$$f_{\rm nt}(r) \equiv \frac{P_{\rm nt}(r)}{P_{\rm tot}(r)} = \frac{P_{\rm nt}(r)}{P_{\rm th}(r)+P_{\rm nt}(r)}, \tag{1.18}$$

This introduces an additional logarithmic gradient accounting for non-thermal pressure within Equation 1.17 and allowing the inferred mass to be interpreted as the true enclosed mass $M_{\rm true}(<r)$. The systematic offset between the $M_{\rm HSE}(<r)$ and $M_{\rm true}(<r)$ is quantified by the *hydrostatic mass bias*:

$$b \equiv 1 - \frac{M_{\rm HSE}}{M_{\rm true}}, \tag{1.19}$$

such that $b > 0$ implies an underestimate of the true mass. Under certain conditions where non-thermal pressure is the only source of bias, and assuming $f_{\rm nt}$ has a modestly varying radial profile, $b \approx f_{\rm nt}$. More generally, b also absorbs contributions from gas temperature inhomogeneities, gas clumping, departures from spherical symmetry, and instrumental calibration uncertainties [Rasia 2014, Pratt 2019]. For example, a cluster with $f_{\rm nt} = 0.2$ at R_{500} and a mild radial gradient, the corrected mass exceeds the purely hydrostatic value by $\sim 25\%$, broadly consistent with the range predicted by numerical simulations [Nelson 2014a, Biffi 2016] and inferred from comparison of X-ray and SZ data [Eckert 2019]. Observationally, the bias is most directly constrained by comparing hydrostatic masses to weak gravitational lensing masses, which are free from assumptions about the gas dynamical state. Such comparisons yield values in the range $1-b \approx 0.74$–0.95, with systematic differences at the $\sim 10\%$ level between different lensing programmes [Mahdavi 2013, Hoekstra 2015, Smith 2016].

The mass bias carries direct consequences for cluster cosmology. Because the cluster mass function is exponentially sensitive to mass near the high-mass end, a systematic underestimate of cluster masses shifts the inferred values of σ_8 and Ω_m [Vikhlinin 2009, Planck XX2014]. The well-known tension between the number counts of Planck SZ-selected clusters and the primary CMB constraints on σ_8 can be partially or fully resolved by invoking a hydrostatic bias of order $b \sim 0.4$–0.6 PlanckXXIV2016, a value substantially larger than simulation predictions. Whether this discrepancy reflects unmodelled non-thermal physics, an underestimate of systematics in either probe, or new physics (e.g. a non-zero summed neutrino mass) remains an open question and is one of the primary motivations for improved cluster mass calibration.

1.5.2 Mass Proxies and Scaling Relations

While continuous mass equations provide physically rigorous mass profiles for individual clusters, they demand spatially resolved spectroscopic data of sufficient quality that their application is necessarily limited to a few hundred well-observed objects at most. Cosmological analyses of large cluster samples instead rely on empirically calibrated *mass proxies*: global observables that correlate with mass and can be measured from lower-quality or unresolved data.

Under the simplifying assumption of self-similar gravitational collapse Kaiser1986, where the internal structure of virialised haloes is universal and determined only by the overdensity at formation, one expects simple power-law scaling relations between the cluster mass M_Δ and global ICM properties. Three mass proxies find particularly widespread application in X-ray and SZ-based surveys.

Mass–X-ray Luminosity

The X-ray bolometric luminosity scales with mass as

$$\frac{M_{500}}{M_0} = A\left(\frac{L_X}{L_0}\right)^{\alpha} [E(z)]^{\beta}, \tag{1.20}$$

where $E(z) = H(z)/H_0$ encodes the self-similar redshift evolution and the expected self-similar slope is $\alpha = 3/4$ for bolometric luminosity. Because $L_X \propto n_e^2 T^{1/2} V$, the luminosity is dominated by dense core gas, making it highly sensitive to the thermodynamic state of the cluster centre. Cool-core clusters are systematically overluminous at fixed mass relative to morphologically disturbed systems, producing a large intrinsic scatter of $\sigma_{\ln M|L_X} \sim 40$–$50\%$ Pratt2009. The principal advantage of L_X is observational accessibility: it can be measured from relatively shallow survey data with modest photon counts, making it practical for large, low signal-to-noise samples such as those from ROSAT and the forthcoming eROSITA all-sky survey.

Mass–X-ray Temperature

The spectroscopic mean temperature is related to mass via

$$\frac{M_{500}}{M_0} = A\left(\frac{T_X}{T_0}\right)^{\alpha} [E(z)]^{\beta}, \tag{1.21}$$

with a self-similar prediction of $\alpha = 3/2$ following from the virial theorem. T_X is less susceptible to core-state bias than L_X because temperature measurements are typically performed in an annular aperture that excises the central cooling region. The resulting scatter is moderate, $\sigma_{\ln M|T_X} \sim 15$–$20\%$ Arnaud2005, though the measurement requires significantly higher photon counts than a luminosity estimate and becomes increasingly uncertain at high redshift where spectral bins must be widened. The hydrostatic bias enters at the calibration stage, since the normalisation A is typically fixed using a reference sample whose masses are derived from the HE equation.

Mass–SZ Signal

The integrated Compton-y parameter, or its X-ray surrogate $Y_X = M_{\rm gas} T_X$ introduced by Kravtsov2006, scales as

$$\frac{M_{500}}{M_0} = A \left(\frac{Y_{\rm SZ}}{Y_0}\right)^{\alpha} [E(z)]^{\beta}, \tag{1.22}$$

with a self-similar slope of $\alpha = 3/5$. The SZ signal is proportional to the total thermal energy of the ICM, $Y_{\rm SZ} \propto \int P_{\rm th}\, dV$, and is therefore a direct in-situ calorimeter of the gravitational potential energy deposited into the gas. Crucially, $Y_{\rm SZ}$ is insensitive to density inhomogeneities (it is linear in n_e rather than quadratic), and the pressure profiles beyond the cluster core are largely governed by the depth of the gravitational potential rather than by the gas physics of the centre. This makes $Y_{\rm SZ}$ the lowest-scatter mass proxy currently available, with $\sigma_{\ln M|Y_{\rm SZ}} \lesssim 10\%$ in ideal conditions daSilva2004, Motl2005, Kravtsov2006, Pike2014. The practical limitations are the sensitivity and angular resolution of SZ instrumentation: at the resolution of *Planck* ($\theta_{\rm FWHM} \sim 5'$), the SZ signal is significantly beam-diluted for clusters at intermediate and high redshift, requiring a clean aperture definition at R_{500} or R_{200} that can be difficult to establish for clusters in complex environments or without prior X-ray data.

Scatter and Bias in Scaling Relations

Each of the above proxies inherits scatter from at least three sources: intrinsic scatter in the physical relation itself (driven by the diversity of cluster formation histories and dynamical states), observational noise, and systematic biases in the mass calibration. Non-gravitational physics — AGN feedback, cooling flows, merger-induced sloshing — breaks the self-similar scaling at varying levels for each observable, modifying the normalisation, slope, and scatter relative to the purely gravitational prediction LeBrun2014, Truong2018. In particular, the slope of the L_X–M and $Y_{\rm SZ}$–M relations is directly affected by the mass-dependent baryon depletion discussed in Section **??**, because both L_X and $Y_{\rm SZ}$ depend on the gas content.

A further systematic that becomes critical in the context of large cluster surveys is Malmquist/Eddington bias: because surveys select preferentially the brightest (most massive) clusters at any given noise level, the observed sample is drawn from the high-scatter tail of the intrinsic distribution. Failure to correct for this selection effect can lead to a $\sim 40\%$ underestimate of the mass at a given survey observable Giles2017. Properly accounting for the survey selection function in the cosmological likelihood is therefore as important as the mass calibration itself.

1.6 Research objectives

This work aims to take an observational approach to the ICM turbulence question. Mainly though the radio SZ angle that probe the large cluster scales. These are the main science questions framed:

- What is the true nature of observable turbulence in the ICM.
- How much of that turbulence is observable with present instrumentation. How much will become observable with near future instrumentation?
- What type of improvements we can we make to remove as much bias as possible from the present methodology of turbulence inference.

The remainder of this thesis is outlined as follows: Chapter 2 looks at the present pipeline framing the current assumptions, Chapter 3 looks at some methods we have done to interrogate those assumptions, Chapter 4 introduces a new method to circumvent some of those biases. Chapter 5 presents results and discussions.

The following publication has been incorporated as Chapter **??**.

1. [**?**] **A. Hamid**, Co-author 1, and Final Author, Non parametric model and simulation of SZ surface brightness, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 2

Assumptions and systematics in turbulence inference from ICM surface brightness fluctuations

> "Real galaxy clusters exhibit a troubling awareness of our assumptions."

The inference of turbulent pressure support from surface brightness data is not a direct measurement — it is the end product of a chain of modeling steps, each of which introduces assumptions about the geometry, thermodynamic state, and statistical properties of the ICM. The difficulty of this methodology is that of error propagation — uncertainties at each stage do not add independently; they compound. This chapter aims to provide a structured examination of the procedural steps, simplifying assumptions, and systematics that constitute the present methodology.

2.1 An end to end overview of the inference chain

Before examining each step in detail, the complete pipeline of how to go from surface brightness to hydrostatic mass bias is first outlined in order to establish the flow of logic and motivate the structure of the discussion that follows. Taken apart the procedure can be summarized into an eight-step process:

1. **Construct the ICM smooth brightness model.** A model of the underlying unperturbed surface brightness distribution is constructed to represent the aggregate structure of the ICM emission. The parametric β-model is the conventional choice though more flexible alternatives exist. This model defines the working boundary between the cluster and turbulent fluctuations hence the accuracy of the smooth model propagates into every downstream step.

2. **Subtract and normalise the fluctuation map.** The smooth brightness model is subtracted from the observed surface brightness image to isolate fluctuations resulting from turbulence. The residual map is typically normalised by the chosen smooth model to produce a fractional fluctuation field. This step attempts to render fluctuations at all radii directly comparable.

3. **Mask and apply a weight map.** Point sources, dusty regions, and other data artefacts are masked in preparation for spectral analysis. A spatial weight map is then applied to define the region of interest, suppress spectral leakage at the map boundaries, and up or down weight different radial zones.

4. **Compute the two-dimensional power spectrum.** A power spectrum as a function of spatial frequency of the fluctuation map is computed. This step characterises the distribution of fluctuation power across spatial scales in the observed POS, and requires additional consideration of the effects of the instrumental beam and noise.

5. **Deproject the power spectrum into three dimensions.** The observed two-dimensional power spectrum is a projection of the underlying three-dimensional fluctuation field. An analytical deprojection kernel is applied to the two-dimensional power spectrum to recover the physical three-dimensional characteristics. This step requires additional models to account for the cluster LOS pressure profile for tSZ and emissivity for X-ray.

6. **Compute the characteristic amplitude spectrum.** The dimensionless characteristic amplitude spectrum is computed from the three-dimensional power spectrum. This quantity represents the RMS fractional density of fluctuations across spatial frequencies, and is the parameter that maps the measured spectra to physical turbulence within the ICM.

7. **Infer the three-dimensional turbulent Mach number.** The peak of the characteristic amplitude spectrum is converted to a three-dimensional volumetric Mach number via a proportional linear relation. The linear relationship is both informative of the thermodynamic state of the ICM while also being dependent on it.

8. **Compute the non-thermal pressure fraction and hydrostatic mass bias.** The three-dimensional Mach number is used to estimate the non-thermal pressure contribution from ICM bulk and turbulent motions. This yields the hydrostatic mass bias: the fractional underestimate of the true cluster mass incurred under assumed conditions of pure hydrostatic equilibrium.

The next sections will elaborate these steps in detail attempting to highlight the biases encoded in each step.

2.2 Smooth brightness model and fluctuations

The first and arguably most consequential step in the pipeline is the construction of a smooth cluster model that separates the large-scale, equilibrium emission of the ICM from the small-scale perturbations that encode turbulent motions. The conceptual foundation for this procedure traces its origin to Reynolds decomposition in fluid mechanics, in which an instantaneous flow quantity is separated into a mean and a fluctuating component [Reynolds ref]. Applied to the observed ICM surface brightness — in both X-ray or tSZ — this decomposition takes the form:

$$S_{\text{obs}}(\boldsymbol{\theta}) = \bar{S}(\boldsymbol{\theta}) + \delta S(\boldsymbol{\theta}), \tag{2.1}$$

where $S_{\text{obs}}(\boldsymbol{\theta})$ is the observed surface brightness at projected position $\boldsymbol{\theta}$, $\bar{S}(\boldsymbol{\theta})$ is the smooth model representing the mean cluster emission, and $\delta S(\boldsymbol{\theta})$ are the residual fluctuations. Physically, $\bar{S}$ encodes the slowly varying emission of the ICM in approximate HE, while δS captures turbulence. The fidelity of $\bar{S}$ therefore determines the amplitude of everything that follows: a model that is too smooth embeds real cluster structure in $\delta S(\theta)$, whereas a model that is too complex absorbs real fluctuations into $\bar{S}$ resulting in underestimates the true fluctuations. This circularity is unavoidable, and sets a limit on systematic uncertainty on the inference of turbulence.

The β-model family

To the first order, $\bar{S}(\theta)$ can be constructed by fitting a parametric model to the observed image. The conventional choice has been the 2D spherical β-model:

$$S(\theta) = S_0 \left[1 + \left(\frac{\theta}{\theta_c} \right)^2 \right]^{-3\beta+1/2}, \tag{2.2}$$

where S_0 is the central surface brightness normalisation, θ_c is the angular radius of the core, and β governs the slope of the outer profile. This functional form closely mirrors the three-dimensional electron density profile in Equation 1.3 of which it is the projected form under the conditions of Equation 1.1. The same functional form can also be applied in the tSZ regime to describe the cluster surface brightness out to larger scales.

Despite its widespread adoption, the single β-model remains inflexible in some cases where it cannot simultaneously reproduce the two component emission of a bright compact core and shallower ambient ICM brightness distribution found in cool core clusters [ref]. Using a single β model for this type of cluster leaves large, structured residuals in the core that bias the fluctuation map at intermediate to large scales [ref], necessitating the double β-model:

$$S(\theta) = S_0 \left[\left(1 + \frac{\theta^2}{\theta_{c,1}^2} \right)^{-3\beta_1+1/2} + R \left(1 + \frac{\theta^2}{\theta_{c,2}^2} \right)^{-3\beta_2+1/2} \right], \tag{2.3}$$

where the parameters $\theta_{c,2}$ and β_2 are for the second core angular radius and slope profile, and R is the relative amplitude. This model, allows each brightness component to be accounted for independently [Mohr et al. 1999; Romero et al. 2024] and is more suited for clusters with excess central emission embedded within an extended envelope.

Granting even a good radial fit, both the single and double β-models carry a further limitation: they both assume spherical symmetry under projection. The observed elongation of clusters also reflects their formation history; clusters undergo the continuous experience of preferential accretion along large-scale filaments, incomplete virialisation following a merger, or tidal interactions with neighboring structures [Jing & Suto 2002; Paz et al. 2006]. When a circular model is applied to an intrinsically elongated cluster, the mismatch between model and data can also produce coherent large-scale residuals such as dipole or quadrupole patterns that bias large scale turbulent power [ref]. An adaptation for this scenario is an elliptical β-model which can be expressed as:

$$S(\boldsymbol{\theta}) = S_0 \left(1 + r_e^2(\theta)\right)^{-3\beta+1/2} + B, \tag{2.4}$$

where

$$r_e^2 = \left(\frac{x_{\mathrm{rot}}}{r_c}\right)^2 + \left(\frac{y_{\mathrm{rot}}}{r_c(1-e)}\right)^2, \tag{2.5}$$

extends the β model to fit the brightness to an elliptical isophotal geometry. Here r_c is the core radius, e is the ellipticity, x_{rot} and y_{rot} are sky-plane coordinates rotated by position angle ϕ from the angular coordinates basis, and B is an additive offset that absorbs any residual background level. (Saxena et al. 2026) adopted this realization as a base ellipse model for their analysis of joint eROSITA and Planck/SPT data of A3266, a merging cluster with a visibly elongated morphology X-ray. However, the final model required an additional patching correction to account for any large scale deviation in elliptical symmetry that the purely elliptical model could not capture. (Heinrich et al. 2024) also warns that the choice of patching scale is not obvious but steps are taken to ensure that whatever scale chosen, is a working (fiducial) sweet spot tuned by comparing the performance of $(\delta\rho/\rho)_k$ power spectrum.

Having explored several variants across the literature, the β model family retains a common limitation irrespective of its specific implementation: the cluster brightness is described by a single globally smooth functional form. Such models have limited freedom to track multiple brightness scales or adapt to localized departures from self-similarity. While additional flexibility can be introduced through this is often achieved at the cost of increased parameter degeneracy and reduced generality, with modifications being tied to specific morphological classes of clusters.

A further complication is the selection of cluster centre is itself not uniquely defined. Depending on the analysis, the adopted centre may correspond to the X-ray brightness peak, the emission centroid, the brightest cluster galaxy, the SZ peak, or the minimum of the gravitational potential. Previous studies have shown a dependency where miscentering by even a few tens of arcseconds generates a dipole residual that mimics injection-scale turbulent power [ref] [ref].

Bias from spherical symmetry

Real galaxy clusters are intrinsically triaxial ellipsoids, the morphology of which is dictated by the underlying dark matter potential well [ref] [ref]. Despite this, a common simplifying assumption made in both simulation and observation is that of spherical symmetry, also referred to as rotational invariance. In observation, the spherical symmetry assumption renders deprojection analytically tractable; in simulation, it is often adopted to reduce computational cost while retaining the large-scale physics of interest. Buote & Humphrey (2012a, b) present a two-paper study investigating the systematic errors introduced into several cluster parameters under a model in which the gravitational potential, rather than the underlying mass distribution, is an ellipsoid of constant shape and orientation.

The studies show that averaging over all possible viewing orientations does not eliminate bias entirely. Although the bias averaged across many cluster facings is small (typically $< 1\%$) for most parameters, the cluster concentration and the integrated Compton-y parameter Y_{SZ}, exhibit the largest orientation-to-orientation scatter. Y_{SZ} especially was found to display a Jekyll and Hyde character reaching $\sigma \sim 10\%$ for Y_{SZ} in the most aspheric cases. This distinction between sample-averaged accuracy and single-object precision makes deviation from spherical symmetry a prime concern for studies of turbulence, that focus on analyzing individual clusters rather than statistical ensembles.

Fluctuation map

With the smooth model defined and the effects of spherical symmetry understood, the residual fluctuation map is obtained by rearranging Equation 2.1:

$$\delta S(\boldsymbol{\theta}) = S_{\text{obs}}(\boldsymbol{\theta}) - \bar{S}(\boldsymbol{\theta}). \tag{2.6}$$

Within the space of the residuals, the key question is to evaluate the size of the fluctuations. This is carried out by taking the RMS amplitude:

$$\sigma_{\delta S}^2 = \langle |\delta S(\boldsymbol{\theta})|^2 \rangle, \tag{2.7}$$

which provides an intuitive but rudimentary statistic, and in an ideal noiseless observation it would reflect the true variance of the ICM surface brightness fluctuations. In practice, the observed residual field contains contributions from multiple sources summed in quadrature:

$$\sigma_{\delta S}^2 = \sigma_{\text{turb}}^2 + \sigma_{\text{noise}}^2 + \sigma_{\text{sys}}^2, \tag{2.8}$$

where σ_{turb}^2 is the variance from genuine ICM fluctuations, σ_{noise}^2 encompasses detector noise, and background fluctuations, and σ_{sys}^2 captures systematic residuals from an imperfectly chosen smooth model. The third term is the most impactful: unlike $\sigma^2_{\rm noise}$ it is correlated with the input smooth model and cannot be reduced by longer source integration time.

The RMS amplitude is therefore a necessary but insufficient diagnostic. While capable of acting goodness of fit measure for $\bar{S}$, its fundamental limitation is that it is a spatially averaged scalar — it collapses information at all scales into a single number. A cluster where fluctuation power is concentrated at ~ 600 kpc scales is indistinguishable from one with identical total variance but power concentrated at ~ 50 kpc, even though these two configurations correspond to entirely different astrophysical scenarios. The RMS can confirm that the residual field is non-trivially large, but it cannot characterise the *nature* of the fluctuations. This is the primary motivation for utilizing the power spectrum, which decomposes the fluctuation variance by spatial scale and allows the injection scale, slope, and amplitude of the turbulent cascade to be resolved independently. rm In practice, the residuals are normalized by the smooth model to acquire the relative fluctuations:

$$\frac{\delta S}{\bar{S}}(\boldsymbol{\theta}) = \frac{S_{\rm obs}(\boldsymbol{\theta}) - \bar{S}(\boldsymbol{\theta})}{\bar{S}(\boldsymbol{\theta})}, \tag{2.9}$$

Aside from recalibrating the fluctuations into a dimensionless scale comparable to the cluster smooth model, the normalization step also has a consequence that can be important for spectral analysis. whereby it compresses the dynamic range of the input fluctuation map. Since $\delta S(\boldsymbol{\theta})$ inherits the surface brightness gradient of the cluster which experiences a drop from the center towards the outskirts, bias from the core can be passed into into the Fourier transform. This is certainly the case in the X-ray regime where brightness tends to drastically drop off by many orders of magnitude moving away from the center but is less impactful in tSZ allowing the normalization to be safely omitted from the analysis if required.

While normalization by $\bar{S}$ can attempt to transform the fluctuations into a field with more uniform variance, it does not eliminate all potential sources of bias. If the model is insufficiently flexible enough to account for them, localized structures such as subgroups and bright radio galaxies can still contribute fluctuations with large amplitudes if left unaccounted for. Consequently, additional spatial weighting and masking procedures are required to suppress contaminated regions prior to spectral analysis.

Masking and apodisation

To further isolate fluctuations from turbulence, a spatial weight map is applied to via multiplication:

$$\delta S_w(\boldsymbol{\theta}) = \omega(\boldsymbol{\theta}) \frac{\delta S}{\bar{S}}(\boldsymbol{\theta}), \tag{2.10}$$

where $\omega(\boldsymbol{\theta}) \in [0,1]$ is the weight map. In its most rudimentary form ω is a binary mask. In practice it performs several functions simultaneously. It is used to exclude contaminated or low signal-to-noise

(SNR) regions, defines the relevant spatial region over which the power spectrum is estimated, and tapers the map boundaries to suppress spectral leakage.

A primary source of unwanted signals in both tSZ and X-ray data are point source contaminants which can inject power across all spatial frequencies. X-ray observations are notorious for a high surface density of resolved point sources originating from AGN. These are typically excised with circular apertures scaled with respect to the local point spread function (PSF). Unresolved sources constitutes a faint, approximately Poissonian background that must be estimated and subtracted from the measured power spectrum separately [ref]. While in tSZ observations the point-source contamination is generally less severe, the Cosmic Infrared Background (CIB) — the integrated, unresolved emission from dusty star-forming galaxies at cosmological distances — constitutes a persistent diffuse contaminant. The effects of the CIB are often minimized in the map making process among other methods [ref].

Beyond the removal of point sources, the weight map defines the spatial region submitted to spectral analysis. It is common practice to restrict the analysis to an annular aperture, excluding both the cluster core and the low signal-to-noise outskirts. The core is excluded because the smooth model is often a poor fit in the innermost regions — where AGN activity, cooling flows, and sharp brightness gradients produce residuals that are not of turbulent origin — and because any model mismatch there would contaminate the fluctuation field at all scales. The outer boundary is set where the surface brightness signal-to-noise falls below a threshold sufficient to recover fluctuations reliably, typically near R_{500} for X-ray observations and somewhat beyond for SZ.

Apodisation. A binary mask with sharp edges is not suitable for Fourier analysis: discontinuities at the mask boundary generate Gibbs ringing that leaks power from large scales into all wavenumber bins, biasing the recovered spectrum. This is mitigated by apodisation — a smooth roll-off that tapers $\omega(\boldsymbol{\theta})$ to zero at both the inner and outer boundaries of the analysis region. The optimal apodisation profile is not universal; it depends on the cluster profile, the analysis scale range, and the signal-to-noise distribution in the specific dataset, and must be validated for each application. Different approaches are used in the literature, ranging from Gaussian tapers to Hanning windows applied in conjunction with the Δ-variance method [Arevalo et al. 2012; Saxena et al. 2026].

An example apodization scheme is seen here:

$$\omega(\boldsymbol{\theta}) = \begin{cases} 0, & |\boldsymbol{\theta}| < r_c, \\ \left[1 - \exp\left(-\frac{(|\boldsymbol{\theta}| - r_c)^2}{\sigma_{\text{in}}^2}\right)\right]\left[1 - \exp\left(-\frac{(R_{200} - |\boldsymbol{\theta}|)^2}{\sigma_{\text{out}}^2}\right)\right], & r_c \le |\boldsymbol{\theta}| \le R_{200}, \\ 0, & |\boldsymbol{\theta}| > R_{200}, \end{cases} \tag{2.11}$$

where r_c is the core exclusion radius, R_{200} is the outer boundary of the analysis region, and σ_{in} and σ_{out} are the roll-off widths at the inner and outer edges respectively. Each factor independently tapers one boundary: the inner factor rises from zero at r_c, the outer factor falls to zero at R_{200}, and their product is unity well within the annulus. The roll-off widths σ_{in} and σ_{out} are free parameters tuned to balance suppression of edge ringing against the loss of usable area at each boundary. For the Coma analysis, we adopt $\sigma_{\mathrm{in}}^2 = 0.02$ and $\sigma_{\mathrm{out}}^2 = 0.18$ [in units — specify], chosen to provide adequate apodisation over the range of scales probed while retaining sufficient usable area within R_{200}.

2.3 Image Power Spectrum

We now turn to the part of the methodology in which *spectral analysis* is applied. The fundamental objective of spectral analysis has remained unchanged since its inception: to decompose a complex, aggregate phenomenon into its constituent, scale-dependent parts. In his foundational experiments with physical optics, Isaac Newton used a glass prism as a hardware-based dispersion mechanism to decompose white light into its discrete constituent colours, unweaving the rainbow into a spectrum ordered by wavelength. In its present-day manifestation, astrophysicists employ numerical Fourier analysis to decompose astrophysical signals into their constituent spatial scales, continuing in this same tradition: where Newton's prism sorted light by wavelength, the Fourier transform sorts a turbulent field by the variance of its fluctuations.

Squared magnitude of Fourier transform is what allows for the signal to be broken down into its . It is the Energy spectral density of the signal. In combination with Fourier theory, Parsevals theorem is to be thanked for this.

$$\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}|f[x,y]|^2 = \frac{1}{M\times N}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}|F[u,v]^2| \tag{2.12}$$

where $\widetilde{\delta}(\mathbf{k})$ is the Fourier transform of the (typically mean-subtracted) field $\delta(\mathbf{x})$, $\mathbf{k}$ is the Fourier wavevector, and N is the area of the observed patch. Averaging $P(\mathbf{k})$ in annuli of constant $|\mathbf{k}|$ yields the azimuthally averaged power spectral density, $P(k)$: a measure of the variance of the field per logarithmic (or linear) bin in spatial frequency. This quantity is the principal observable used throughout this work to characterise the statistical properties of ICM pressure and density fluctuations.

In practice, several distinct estimators are used to compute $P(k)$, each suited to particular data characteristics (sky coverage, masking, signal-to-noise regime) and each carrying its own trade-offs in bias, variance, and robustness to incomplete or irregularly sampled data. The estimators most commonly applied in the study of astrophysical and ICM turbulence are summarised below.

- **Periodogram.** The direct FFT-based estimator of Equation **??** is the classical and most widely accessible method, and is computationally the cheapest. However, the standard periodogram assumes a regularly sampled, gap-free grid; it is therefore not robust to missing data, irregular sampling, or masked regions. For irregularly sampled time series or maps with significant masking, the **Lomb–Scargle periodogram** [ref] is frequently substituted, as it generalises the discrete Fourier transform to non-uniform sampling by fitting sinusoids directly to the unevenly spaced data rather than relying on an implicit regular grid.

- **Correlogram.** This method applies the FFT to the autocorrelation function (ACF) of the field rather than to the field itself, as prescribed by the Wiener–Khinchin theorem, which states that the power spectrum and the autocorrelation function form a Fourier transform pair. Because the ACF can be estimated directly from the available data pairs without requiring a complete regular grid, this approach is comparatively more tolerant of missing data than the direct periodogram.

Its principal drawback is the need to apply and tune an apodising window function to the ACF before transforming, in order to suppress the spectral leakage and ringing that would otherwise result from the implicit truncation of the correlation function at large lags.

- **Flat-sky approximation.** For observations subtending a sufficiently small solid angle on the sky (typically $\theta \lesssim 9^\circ$ across), the curvature of the celestial sphere can be neglected and the observed patch treated as a 2D Euclidean plane. This greatly simplifies the computation of the power spectrum, since the spherical-harmonic decomposition required for full-sky analysis (as used, e.g., for CMB and large-area SZ maps) can be replaced by a standard planar Fourier transform. This approximation is well suited to single-pointing or mosaicked cluster observations, such as those from *XMM-Newton*, *Chandra*, or ground-based SZ interferometers and bolometer arrays, where the field of view is at most a few tens of arcminutes.

- **Δ-variance.** Rather than decomposing the signal into a discrete set of Fourier modes, the Δ-variance method Stutzki1998, Ossenkopf2008 operates in scale space: the data are filtered with a localised kernel (most commonly a Mexican-hat or French-hat wavelet) tuned to isolate fluctuations at a specific spatial scale ℓ, and the variance of the filtered map is computed as a function of ℓ. This naturally handles irregular masks and gaps, since the filtering operation can be restricted to the unmasked pixels. A widely used extension, the Arevalo2012 method, refines the choice of filter to better approximate a true band-pass in Fourier space. The principal weakness of Δ-variance-type estimators is that they introduce a systematic bias when applied to steep power-law signals, because the filter kernels have non-zero response outside their nominal target scale, causing power to leak between adjacent scale bins; this bias must be explicitly modelled and corrected for, particularly for ICM pressure or density power spectra that are expected to follow a Kolmogorov-like power law over an extended inertial range.

An honourable mention should also be made of **structure function** methods, which characterise turbulence through the second-order moment of the field increments, $\langle[\delta(\mathbf{x}+\mathbf{r})-\delta(\mathbf{x})]^2\rangle$, as a function of separation r. The structure function and the power spectrum are mathematically related (for a stationary, statistically homogeneous field they form a Fourier pair, up to the relevant moment order), but the structure function is computed in real space and can therefore be more directly tolerant of complex survey geometries; it does, however, mix power from a broad range of scales into each value of r, making the recovery of a localised spectral index less direct than with a true power spectral estimator.

Moving forward, for the remainder of this work we adopt the direct periodogram (Equation **??**) as our default power spectrum estimator, unless explicitly stated otherwise.

2.3.1 Instrumental Beam Effects

Whenever the sky is observed with a real telescope, a finite degree of blurring is unavoidably applied to the resulting image. This effect is well approximated as a convolution of the true sky signal with the instrument's point spread function (PSF), which for most radio, mm-wave, and X-ray telescopes is itself well approximated by a 2D Gaussian of width set by the beam full width at half maximum (FWHM). Because convolution in real space corresponds to multiplication in Fourier space, the observed power spectrum is related to the true underlying power spectrum by a purely multiplicative transfer function:

$$P_{\text{obs}}(k_{2D}) = B(k_{2D})^2 H(k_{2D})^2 P_{\delta y/\bar{y}}(k_{2D}), \tag{2.13}$$

2.4 Deprojection and 3D reconstruction

Becasue we are analyzing surface brightness everything we see is a two dimensional deprojection of a three dimensional phenomena. This invariable leads to some of information. Projection is simply an integral.

There is an Abel transfrom used for

2.5 Physical Interpretation

Normalization (amplitude A_δ) of the X-ray density power spectrum is linearly related to the turbulent Mach number. The ratio of density to Pressure is 1 which is pretty remarkable. However this is conditional on the thermodynamic state.

You have multi scale, multi layer, multi mach numbers. How does one even know what the injection scale really is? Here is some theory at least one page to prove it. Suffice to say that shit is complex.

Kolmogorov turbulence [ref] applies in the subsonic, weakly compressible limit ($\mathcal{M} \ll 1$) where density fluctuations are small and the $E(k) \propto k^{-5/3}$ spectrum holds. For the bulk ICM with $\mathcal{M}_{3D} \lesssim 0.5$ this is a reasonable zeroth-order approximation [ref].

Burgers turbulence [ref] applies in the supersonic, shock-dominated limit ($\mathcal{M} \gg 1$) where the flow is dominated by discontinuous shocks rather than smooth eddies. The energy spectrum steepens to $E(k) \propto k^{-2}$. While the bulk ICM is subsonic, merger-driven shocks can locally reach this regime, particularly at cluster outskirts and along merger axes.

Iroshnikov–Kraichnan (IK) turbulence [ref] generalises the Kolmogorov framework to isotropic MHD. The presence of a mean magnetic field introduces Alfvén wave propagation as an additional timescale competing with the eddy turnover time, modifying the energy spectrum to $E(k) \propto k^{-3/2}$. Given that the ICM is magnetised at the level of ~ 1–$10\,\mu$G, MHD effects cannot be neglected entirely.

Goldreich–Sridhar (GS) turbulence [ref] extends the MHD description to the anisotropic case, where turbulent eddies are elongated along the local magnetic field direction. In this framework the cascade is critically balanced between the eddy turnover time and the Alfvén wave period, and the energy spectrum along and perpendicular to the field differs. This anisotropy is particularly relevant for the interpretation of ICM turbulence in the presence of stratification and ordered magnetic field geometries.

PITSZI:

$$P_{3D}(k) = \sigma_P^2 \frac{k_{3D}^2 \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right)}{\int 4\pi k_{3D}^{\alpha+2} \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right) dk_{3D}} \quad (2.14)$$

$$P_{3D}(k) = \sigma_P^2 \frac{k_{3D}^2 \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right)}{\int 4\pi k_{3D}^{\alpha+2} \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right) dk_{3D}} \quad (2.15)$$

2.6 Propagation and Compounding of Systematic Uncertainties

As you can see is heavily reliant on inputs and assumptions. Is this a summary?

Hello I am here. The following publication has been incorporated as Chapter 3.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 3

Methodologies for Testing Turbulence Recoverability

"Doors and corners. Thats where they get you."

In this chapter we talk about how we interrogated all the assumptions that go into the pipeline.

3.1 Non Parametric Methods

Without any prior assumptions on turbulent mach numbers.

3.1.1 Iso Contour Surface Brightness

Its a type of harmonics based algorithm. It interpolates concentric ellipses.

3.1.2 Fourier Mode Smooth models

This is actually a Low Pass Filtering

3.1.3 Ellipse models

This is actually two contributions in one. One is a sniper genetic algorithm method based method. And the other is a drone swarm.

Genetic Algorithms with Differential Evolution

Bayesian Inference with emcee

Probably some corner plots here. Talk about the Likelihood function.

3.2 fBm Fields and Artificial Clusters

This is where you say that the field of cosmology is really over reliant on simulations. Because samples are small. Viewing is expensive and even when we can view are limited of what what we can see.

3.2.1 GCMC cluster

3 knob fluctuation experiment

Hello I am here. The following publication has been incorporated as Chapter 4.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 4

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about the assumptions.

4.1 Introduction

Add your text here.

Chapter 5

Conclusion

Conclude your thesis.

Appendix A

Appendix

Write your appendix here. Following two are examples.

A.1 Name of Appendix-1

A.2 Name of Appendix-2

Appendix B

Example of Citations

This text is only for Bibliography testing purposes. This is a book by Cawvey et al. [?]. There are few journal articles [?, ?] in this bibliography. This is a proceeding paper [?]. This is a PhD thesis by Peerling [?]. These are few other types of citations [?].

Appendix C

Example of Code

C.1 Find the greatest number from a list of numbers in *Python*

```
a=[1,2,3,4,6,7,99,88,999]
    max= 0
    for i in a:
        if i > max:
            max=i
    print(max)
```

Appendix D

Example of Equations

$$E = mc^2 \tag{D.1}$$

$$a = b + c \tag{D.2}$$

$$x = y + z \tag{D.3}$$

$$\begin{aligned} a &= b + c \\ &\quad + d + e \end{aligned}$$

$$sin x + cos x = 1 \tag{D.4}$$

Appendix E

Example of Figures

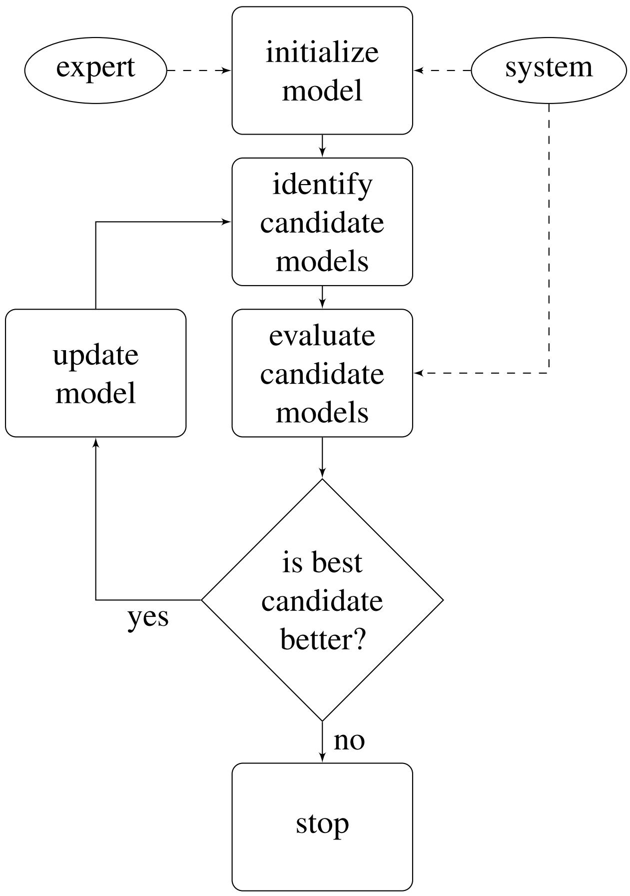
Figure E.1: The University Of Queensland

The Figure E.1 represents beauty of the UQ campus.

Appendix F

Example of Flow Charts

Figure F.1: Flow Chart



Flow chart F.1 is a simple example.

Appendix G

Example of Tables

Here is a really simple table G.1.

Table G.1: Name of the Australian Cities

Number	**Name**
1	Brisbane
2	Sydney
3	Melbourne
4	Canberra
5	Perth
6	Adelaide
7	Hobart
8	Darwin

Endquote goes here.

Author of quote,
Source of quote